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Launching an Application on AWS Beanstalk

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AWS Elastic Beanstalk is a fully managed service that simplifies hosting web applications by automatically provisioning and managing the underlying infrastructure. Developers provide their code, and Elastic Beanstalk handles the rest.

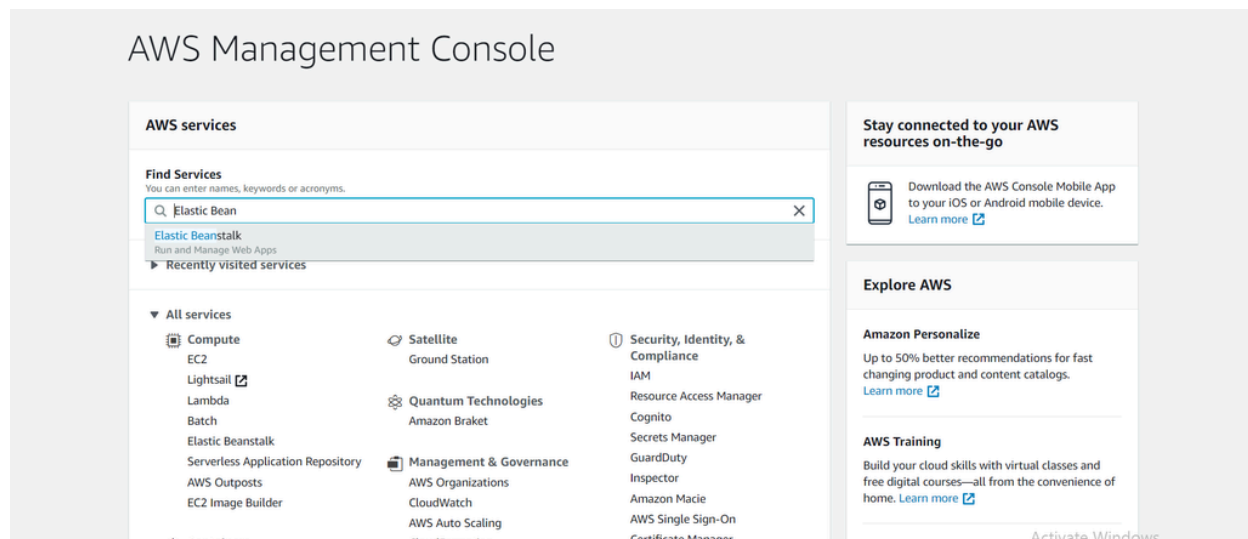
- Virtual machines, load balancers, operating systems, language runtimes, and web servers are all provisioned automatically.
- Supports Java, .NET, PHP, Node.js, Python, Ruby, and Go.
- Runs on Apache, Nginx, Passenger, and IIS depending on the platform.
- Elastic Beanstalk itself has no additional charge — you pay only for the underlying AWS resources it provisions.
- A single application can have multiple environments such as dev, staging, and prod running simultaneously.

Core Elastic Beanstalk Concepts

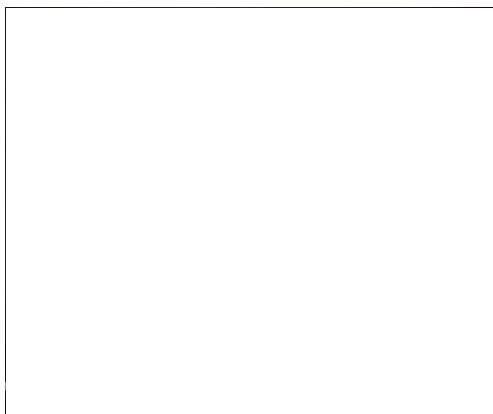
- **Application:** A logical container for your entire project. Think of it as a folder that holds multiple versions and environments.
- **Environment:** The actual running instance of your application — the collection of AWS resources (EC2, Load Balancer, Auto Scaling Group, etc.) that Elastic Beanstalk creates and manages on your behalf.
- **Application Version:** A specific packaged snapshot of your code, typically a .zip or .war file.
- **Platform:** The full technology stack your application runs on, combining the operating system, language runtime, web server, and application server (e.g., Python on Amazon Linux 2 with Gunicorn, or Go on Amazon Linux 2).

Launching a Sample Application

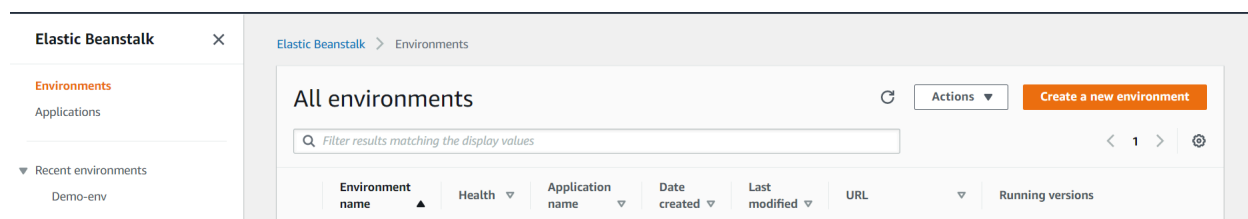
Step 1: Log in to your AWS account, search for Elastic Beanstalk in the main search bar, and select it to open the service dashboard.



Step 2: Create a New Application



- Click Create application from the dashboard.
- In the configuration wizard, select Web server environment as the environment tier.
- Under Application information, enter a name such as my-first-gfg-app.



Step 3: Configure the Environment

- Leave the Environment name as auto-generated, or customize it.
- Leave the Domain field blank for an auto-generated subdomain, or enter a preferred value.
- Under Platform, select your technology stack from the dropdown (e.g., Python). Elastic Beanstalk will pre-install the appropriate runtime and web server.

Elastic Beanstalk

Environments

Applications

Application name

demoGFG

Up to 100 Unicode characters, not including forward slash (/).

Application tags (optional)

Environment information

Choose the name, subdomain, and description for your environment. These cannot be changed later.

Environment name

Demogfg-env

Domain

Leave blank for autogenerated value

.us-east-1.elasticbeanstalk.

Check availability

Description

First elastic beanstalk application

Platforms published and maintained by AWS Elastic Beanstalk. [Learn more](#)

Platforms created and owned by you

Platform

PHP

Platform branch

PHP 7.4 running on 64bit Amazon Linux 2

Platform version

3.1.1 (Recommended)

Application code

☒ Sample application
 Get started right away with sample code.

☐ Existing version
 Application versions that you have uploaded for demoGFG.

-- Choose a version --

☐ Upload your code
 Upload a source bundle from your computer or copy one from Amazon S3.

Cancel

Configure more options

Create environment

Step 4: Select Sample application to deploy a pre-built application provided by AWS. This is the recommended option for first-time setup, as it is guaranteed to work and confirms that your environment is configured correctly.

Application code

- ☐ Sample application
 Get started right away with sample code.
- ☐ Existing version
 Application versions that you have uploaded for demoGFG.

-- Choose a version --
- ☒ Upload your code
 Upload a source bundle from your computer or copy one from Amazon S3.

Version label

Source code origin

Maximum size 512 MB

- ☒ Local file
- ☐ Public S3 URL

Choose file

☐ No file uploaded

► Application code tags

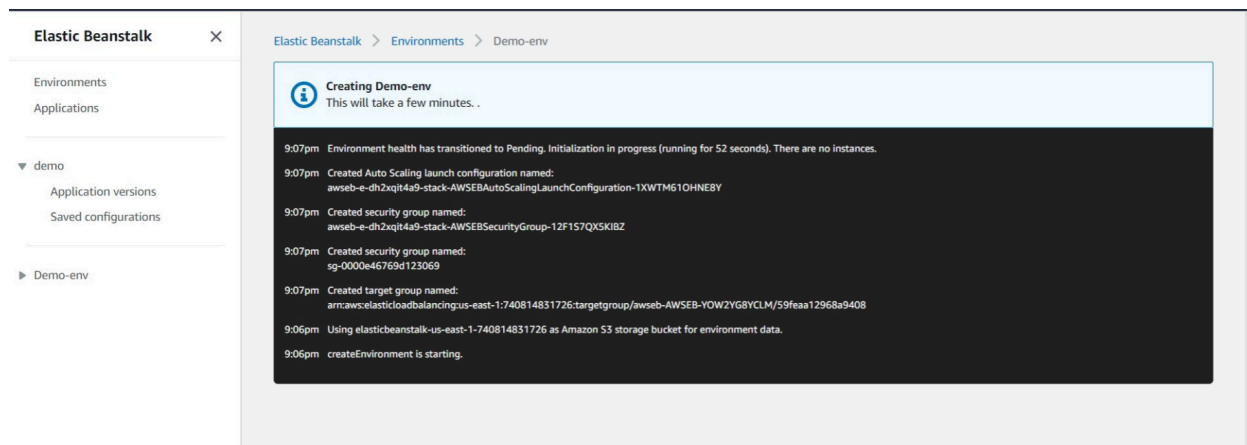
Cancel

Configure more options

Create environment

Step 5: Review and Launch

- Click Next to skip optional advanced configuration pages.
- On the final review page, acknowledge the service role creation required for Elastic Beanstalk to manage resources on your behalf.
- Click Submit.



Step 6: You will be taken to the environment launch dashboard. Elastic Beanstalk provisions a complete infrastructure stack in the background. The event log will display each resource as it is created:

- **S3 Bucket:** Stores your application versions.
- **Security Group:** Acts as a firewall controlling instance traffic.
- **EC2 Instance:** The virtual server running your application.
- **Auto Scaling Group:** Manages instance count based on load.
- **Elastic Load Balancer:** Distributes incoming traffic across instances.
- **CloudWatch Alarms:** Monitors health and performance metrics.

***Note:** This process typically takes 5–10 minutes to complete.*

Demogfg-env

☐ Refresh
 Actions

[Demogfg-env.eba-mmybpjyc.us-east-1.elasticbeanstalk.com](#) (e-dz7gkw3rqc)

 Application name: **demoGFG**

Health

Degraded

Causes

Running version

Sample Application

Upload and deploy

Platform

PHP 7.4 running on 64bit Amazon Linux 2/3.1.1

Change

Recent events

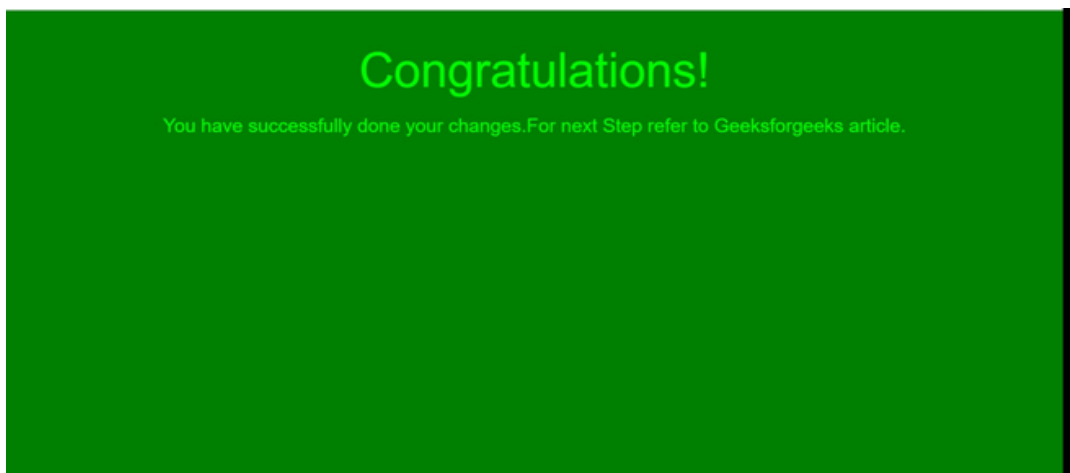
Show all

< 1 >

Time	Type	Details
2020-09-28 00:41:50 UTC+0530	WARN	Environment health has transitioned from Info to Degraded. Command failed on all instances. Incorrect application version found on all instances. Expected version "Sample Application" (deployment 1). Application update is aborting. 1 out of 1 instance completed (running for 52 seconds).

Step 7: Access Your Live Application

- Once launch completes, the health status turns green with an "Ok" status.
- Click the Domain URL displayed at the top of the dashboard (e.g., my-first-gfg-app-env.eba-xyz.us-east-1.elasticbeanstalk.com).
- Your live sample application will open in a new browser tab.



Cleaning Up

Elastic Beanstalk is free, but the AWS resources it provisions (such as EC2 instances) incur charges. Always terminate your environment after practice.

- Navigate to your environment's dashboard.
- Click the Actions dropdown in the top-right corner.
- Select Terminate environment.
- Confirm by typing the environment name and clicking Terminate.

Note: Terminating the environment deletes all associated AWS resources that were provisioned during creation.

Comment

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